Skill, Agent, or Neither?

Five Orbitly support capabilities, each decided, built and tested on its own

Saadurya | 2026-10-03 | SaaS Project: building skills and agents for customer support

1. The take

Each capability goes where its failure can be caught: format in skills, volume in agents, risk in code.

The deciding question is not whether an LLM can do the job. It is what a miss costs and what would catch it. Where a miss is expensive and the rule is mechanical, the work is code. Where it is a repeated format built from one or two documents, it is a skill. Where it is volume across records or systems, it is an agent with its own context and read-only tools. Anthropic's documentation draws the same line: instructions in CLAUDE.md or a skill are requests, not guarantees, and a rule that must hold every time belongs in code (see Sources).

Layer	Holds	Capabilities	Evidence from the tests
Code	Risk: a costly miss, mechanical rule	1 Triage and escalation (Neither); `check_figures.py` inside capability 2	The script rejected a draft that repeated the customer's "40%" before it could reach a reply. Keyword escalation costs no model tokens.
Skill	Format: a known output from one or two documents	2 Replies; 5 Bug reports	Policy window changed from 30 to 45 days with the skill text untouched, and the reply flipped from refusal to refund.
Agent	Volume: many records or systems, raw material kept out of the main session	3 At-risk briefing; 4 Weekly QA	Three systems reconciled into a 375-word briefing with no data from the look-alike customer. Eight tickets returned as one scorecard.

Outcome: 2 Skills, 2 Agents, 1 Neither, all built and tested standalone, with no orchestration, coordinator or pipeline anywhere in the project.

#	Capability	Decision	Built as	Why
1	Triage new tickets: categorise, route within minutes, escalate outages and legal or security keywords immediately	**NEITHER**	Ticketing-tool automation (documented, not built)	Keyword rules and a routing table fire in seconds and guarantee recall on legal and security terms. A person-invoked skill or agent is the wrong shape for an event-driven, latency-critical, costly-to-miss step.
2	Replies in company tone; never state a refund or pricing figure that is not in the current policy	**SKILL**	`reply-drafter` + `check_figures.py`	One repeated, format-driven step. Fast-changing facts stay in `docs/policy.md` and are read fresh each run, so nothing goes stale inside the skill.
3	At-risk customer: pull billing, usage and past tickets into a one-page briefing	**AGENT**	`at-risk-briefing` (read-only)	Three systems, steps that vary per customer, and reconciliation across sources. The raw logs stay in the agent's own context; only the briefing returns.
4	Weekly grading of a rolling sample of closed tickets plus a failure-pattern summary	**AGENT**	`ticket-quality-review` (read-only)	A batch job: read standards, grade every ticket, count, mine patterns. The bulk ticket text stays out of the main session; only the report returns.
5	Bug report: check the changelog for a known issue, otherwise write it up for engineering with repro steps	**SKILL**	`bug-report-writer`	Two bounded steps: look up one document, fill a fixed template. Predictable output and no open-ended tool use.

2. How each decision was made

Three questions, asked in order of every capability:

Question	If yes
Is the step structured, predictable and deterministic (rules, keywords, routing, de-duplication)?	Use **code or existing automation**. Spend no LLM tokens on it: extraction and de-duplication are code; LLMs are for context, interpretation and generation.
Is the output a known, reusable format and the task lightweight and repetitive?	Use a **Skill**: a reusable instruction package, lighter than an agent.
Is it goal-driven, multi-step and tool- or source-heavy, with a workflow that varies per case, or does it need heavy raw material kept out of the main context?	Use an **Agent** (or sub-agent): its own context window, returning only the result.

The two closest calls

Capability 4 is an agent, not a skill. Grading against a rubric is structured and skill-like, but the job is volume: many tickets read, graded, counted and pattern-mined. That work belongs in a separate context that returns only the report. The rubric and standards live in `docs/`, so the reusable format is still reusable.

Capability 1 is Neither. A missed legal or security keyword is costly, so deterministic recall beats probabilistic judgement. A cheap LLM classifier could later sit inside the ticketing automation as an optional step, but that is an API call in a workflow, not a Skill or Agent in a person's session.

3. What was built

Item	Type	Location	Scope and design choices
reply-drafter	Skill	`.claude/skills/reply-drafter/`	Reads `docs/policy.md` and `docs/tone-of-voice.md` every run. Only policy figures, no derived amounts, no promises, hand-offs described as the writer's own action. Ships `scripts/check_figures.py`, which mechanically rejects any money, percent or duration figure not in the policy, banned phrases and leaked section numbers. Tools: Read, Write, Edit, Bash(python3 only).
bug-report-writer	Skill	`.claude/skills/bug-report-writer/`	Reads `docs/changelog.md` (fixes and open known issues). Classifies KNOWN-FIXED, POSSIBLE-REGRESSION, KNOWN-OPEN or NEW; similar-sounding entries are cited as 'related but different'. Never invents repro steps; gaps go under NEEDS INFO. Tools: Read, Grep.
at-risk-briefing	Agent	`.claude/agents/at-risk-briefing.md`	Tools **Read, Grep, Glob only**. One customer, always filtered by `customer_id` because names can collide. Computes utilisation, trend and days to renewal; separates source-tagged facts from inference; one-page structural limits; no promises or discounts.
ticket-quality-review	Agent	`.claude/agents/ticket-quality-review.md`	Tools **Read, Grep, Glob only**. Reads rubric, policy and tone guide fresh each run; scores A-E 0/1/2 with quoted evidence; auto-fail on any policy violation; a pattern needs 2+ tickets; deterministic sampling rule above 15 tickets.
Project context	Config	`CLAUDE.md`, `.claude/rules/`	Compact always-loaded overview (under 200 lines) plus a path-scoped rule that loads only for `drafts/**` and `*-reply.md`.

Course concepts applied

Concept	Where it shows up
CLAUDE.md is a concise, always-loaded overview: under ~200 lines, no secrets	`CLAUDE.md` holds layout, conventions and guardrails only; no keys, no procedures.
Policy lives in CLAUDE.md, enforcement lives in code	Guardrails are stated in `CLAUDE.md` and enforced by `check_figures.py` and by agents that have no write tools. Real access control would sit in billing and ticketing, not in prompts.
.claude/rules for conditional instructions	`customer-facing-writing.md` is scoped by path, so it loads only when drafts are touched and `CLAUDE.md` stays short.
Project-level vs global; skills in `.claude/skills/<name>/SKILL.md`	Everything is project-level and applies only to this repo; each skill has its own folder and `SKILL.md`.
Agent selection is by description	Each agent description names its trigger phrases, scope and input, and says it is read-only.

Concept	Where it shows up
Sub-agents have isolated context and return only the result	Both agents live in `.claude/agents/`; raw logs and ticket text stay out of the caller's context.
LLM for context and generation, code for deterministic steps	Capability 1 is Neither; figure and banned-phrase checks are a script; the verifier recomputes arithmetic in code.
Traceability: every number traceable	Briefings tag each fact with system and record id; QA scores quote ticket text; each run's tokens and cost are saved as JSON.
Token cost, model choice, scope creep	Section 6: measured cost per run and the model-choice pick.
One feature per session	Each skill and agent is tested in a fresh isolated copy of the project, one per run.

4. Standalone tests: inputs and outputs

Each component ran on its own through the Claude Code CLI in headless mode (`claude -p`; skills via the Skill tool, agents via `--agent`; CLI 2.1.288, claude-sonnet-5-5), inside a fresh copy of the project that excludes the `tests/` folder, so the answer key was never visible. The runner is `tests/run_tests.sh` and outputs are in `tests/*/output_*.md`. Outputs are LLM-generated and vary slightly between runs; the checks in section 5 are deterministic and re-runnable. Each test also has a screenshot from a second, live run in Claude Code (VS Code terminal, interactive mode) showing the output exactly as the support employee receives it, in plain language rather than JSON or code. Wording and counts differ slightly between the two runs, and the notes under each screenshot say where.

4.1 Skill: reply-drafter

Test 1 input (T-70231): refund request 41 days after an annual purchase, plus downgrade, competitor price match and Enterprise pricing

Ticket: T-70231
Today's date: 2026-10-03
Sending agent: Sam
Account context: Pro plan, annual, 10 seats, purchased 2026-08-23 (first purchase, no earlier subscription).

Customer message (from Dana Whitfield, ops lead):

Hi, we bought the annual Pro plan for 10 seats about six weeks ago and honestly only 4 of us ended up using it.
I'd like a full refund of the $4,900 please. If you can't do that, what would we pay if we dropped to the Starter plan?
Also a competitor offered us 40% off, can you match that? And what would Enterprise cost for 25 people?
We need an answer by tomorrow.

Test 1 output (policy v2026.10: annual refund window 30 days)

Reply draft

Hi Dana,

I can't offer a full cash refund on this one. Annual plans are refundable in full within 30 days of the initial purchase, and your purchase on 2026-08-23 is now past that window.

What I can do is ask Billing to check whether an account credit is possible for unused full months left on your annual term. Billing calculates any credit and shows it to you before anything is confirmed, so I can't give you a figure myself.

On Starter: the list price is $190 per seat per year. A downgrade takes effect at the start of your next billing cycle, and the price difference isn't refunded. Billing can work out what that looks like for your team.

I can't match the competitor's offer. We have no discounts beyond the ones in our standard pricing, and I can't promise one. Enterprise pricing is custom, so I don't have a number for 25 people. I'm passing both questions to Sales so they can speak with you directly.

I know you wanted an answer by tomorrow. I can't promise a timeline for Billing or Sales, but I'm sending both requests today and I'll update you here as soon as I hear back.

Sam
Orbitly Support

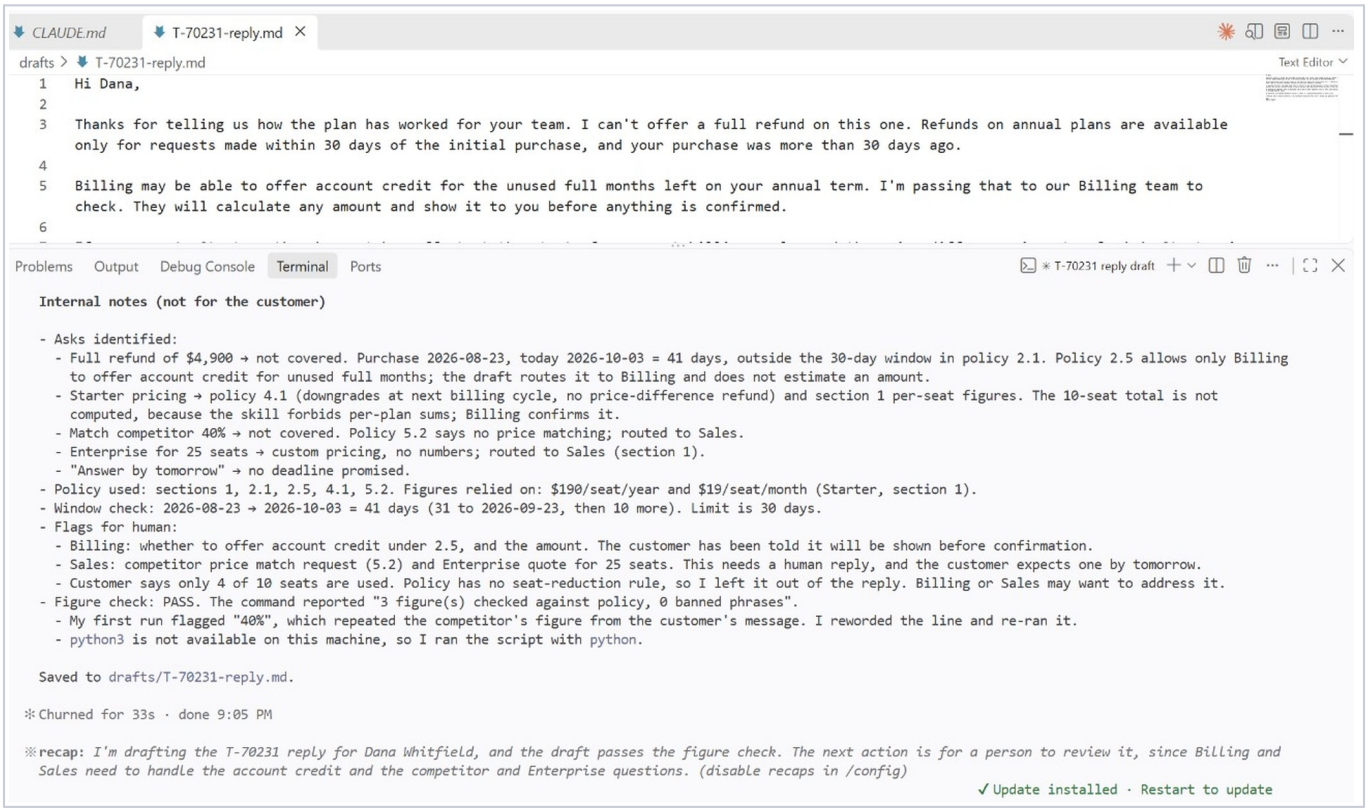
Internal notes (not for the customer)

- **Asks identified:**
- Full refund of $4,900: not covered, because it is outside the refund window. Possible account credit goes to Billing.
- Starter downgrade cost: partly covered. The per-seat list price and the downgrade timing are in policy. The total goes to Billing.
- Match the 40% competitor offer: not covered, route to Sales.
- Enterprise price for 25 seats: not covered, route to Sales. No figures quoted.
- Answer by tomorrow: no deadline promised.
- **Policy used:**
- 2.1: full refund within 30 days of the initial purchase.
- 2.5 and 2.6: no cash refund outside the window. Billing may offer credit for unused full months, and Billing calculates and shows it.
- 1: Starter annual is $190 per seat per year. Enterprise is custom, so support quotes no number.
- 4.1: a downgrade takes effect at the next billing cycle, with no refund of the price difference.
- 5.2: no competitor price matching or custom discounts.
- **Window check:** Purchased 2026-08-23, today 2026-10-03. That is 41 days (31 days to 2026-09-23, plus 10), which is 11 days past the 30-day window. The customer's "about six weeks" matches. This is a first purchase, so the renewal window (2.2) doesn't apply.
- **Flags for human:**

- Billing Lead: needs to decide whether to offer account credit under 2.5, and Billing needs to calculate any amount. Any one-off refund exception would also need Billing Lead approval (section 6).
- Sales: needs to take the competitor discount request and the Enterprise inquiry for 25 seats.
- The customer wants an answer by tomorrow (2026-10-04). Nothing in policy commits us to that, so a human should decide whether to chase Billing and Sales today.
- I did not repeat the customer's $4,900 figure or compute any Starter total.
- **Figure check:** PASS: 2 figure(s) checked against policy, 0 banned phrases. The draft is saved at `drafts/T-70231-reply.md`.

Result: declines the refund (41 days against a 30-day window), offers only what policy allows (Billing credit review, downgrade at the next cycle), refuses the competitor match, quotes no Enterprise price and does not repeat the customer's $4,900 or 40%. Figure check passed.

Screenshot, live run: the draft reply opens in the editor (top) and the internal notes for the human reviewer print in the terminal (not for the customer)



In the live run the skill's figure checker caught the customer's own '40%' echoed in the first draft; the draft was reworded and re-checked, ending with 3 figures checked against policy and 0 banned phrases. The agent also noted that `python3` was not available on this Windows machine and ran the script with `python`, so the check does not depend on one interpreter name.

Test 2 output: same input, policy changed to a 45-day window

Reply draft

Hi Dana,

Yes, you can have a full refund. Annual plans are fully refundable if requested within 45 days of the initial purchase, and your request is inside that window. I'm sending it to Billing now. They'll calculate the refund and show it to you before anything is confirmed.

If you'd rather stay on a smaller plan, Starter is $190 per seat per year. A downgrade takes effect at the start of your next billing cycle and doesn't refund the price difference. Billing can walk you through the options and show you any figures before you decide.

On the competitor offer, I can't match competitor prices or give custom discounts. I'm passing that request to Sales so they can review it with you.

For Enterprise, we don't quote pricing through Support. I'm also passing your 25-person question to Sales.

I can't promise exact timing for Sales or Billing, but I've flagged that you need an answer by tomorrow. I'll follow up with you as soon as I have an update.

Sam
Orbitly Support

Internal notes (not for the customer)

- Asks identified:
- Full refund of Pro annual -> covered by policy (within window). Billing calculates and confirms the amount.
- Starter price if they downgrade -> covered (list price). Downgrade applies next cycle with no refund of the difference. Billing shows any totals.
- Match the competitor's 40% -> not covered. Route to Sales.
- Enterprise for 25 seats -> no figures from Support. Route to Sales.
- Answer by tomorrow -> no deadline promised. Flagged below.
- Policy used: §2.1 (45-day full refund on annual plans), §2.6 (Support never quotes refund amounts), §1 (Starter $190 per seat per year; Enterprise custom, route to Sales), §4.1 (downgrades), §5.2 (no matching or custom discounts).
- Window check: Purchase 2026-08-23 to today 2026-10-03 is 41 days (31 days to 09-23, plus 10 more). The 45-day window ends 2026-10-07, so 4 days remain. The customer said "about six weeks", which is roughly right (41 days is about 5.9 weeks). The refund request is in the window, so request it now to avoid the cutoff.
- Flags for human:
- Sam: send the refund request to Billing today. The window closes 2026-10-07, so this is time-sensitive.
- Sales needs to handle the competitor-match request and the Enterprise quote for 25 seats.
- The reply says the "by tomorrow" deadline is flagged, but no response time is promised. Make sure Billing and Sales are actually told.
- The customer may refund and then rebuy on Starter instead of downgrading. That is a Billing decision, and the reply doesn't offer it.
- The reply says I'm "sending it to Billing now." Make sure that handoff actually happens before sending.
- Figure check: PASS. `PASS: 2 figure(s) checked against policy, 0 banned phrases`. The first run failed on "40%", the customer's own figure. I changed it to "the competitor offer" and the check passed on the second run.

Draft saved to `drafts/T-70231-reply.md`.

Result: with only the policy file changed, the reply flips to a full refund citing 45 days. Figures are read live, not stored in the skill.

4.2 Skill: bug-report-writer

Test A input: a bug already fixed in a newer release

Customer report (ticket T-71002, Pro plan, workspace "Kiwi Clinics"):

Hi, since Monday our dashboards show yesterday's date when we filter by "Today". We're in Auckland (UTC+13 at the moment). Everyone on the team sees it. We're on version 4.11.2 of the web app. Chrome and Safari both do it.
Reports built on those dashboards are off by a day, which is confusing for our managers.

Test A output

Classification: KNOWN-FIXED

Match: OB-3391, fixed in 4.12.3 (2026-09-18) Related-but-different: OB-3310
Reasoning: The customer's dashboard "Today" filter shows yesterday's date in Auckland (UTC+13). That matches OB-3391, which fixed dashboard date filters showing the previous day for workspaces in UTC+12 and above. They are on 4.11.2, older than 4.12.3, so the fix applies to them. OB-3310 (saved filters not applied on shared dashboard links, 4.12.2) is related but different: it has a different symptom and trigger, and it has nothing to do with time zones.

Suggested customer note

Hi Kiwi Clinics team, thanks for the detailed report, and sorry for the confusion with your manager reports. This is a known issue: dashboard date filters, including "Today", showed the previous day for workspaces in time zones UTC+12 and above, which includes Auckland at UTC+13. It was fixed in version 4.12.3, released on 2026-09-18. You're on 4.11.2, so updating to 4.12.3 or later should resolve it. If "Today" still shows yesterday's date after you update, please reply with the version number you see and we'll look into it further.

Notes for the support agent

- The customer says it started "since Monday", but the changelog doesn't say why a 4.11.2 workspace would only start showing this now. Don't tell them what triggered it.
- The changelog doesn't say how web app updates reach customers (automatic or manual). Check how Kiwi Clinics gets the update before telling them to "update".

- Reports already built on the affected dashboards may hold previous-day figures. The changelog doesn't say whether the fix corrects them, so don't promise it does.

Result: KNOWN-FIXED, matched to OB-3391 (fixed in 4.12.3); OB-3310 cited as related but different; the customer's version gap stated.

Test B input: a genuinely new bug with a similar-sounding changelog entry as a distractor (OB-3310)

Customer report (ticket T-71044, Enterprise plan, workspace "Acme Production"):

This morning our admin renamed the workspace from "Acme-Prod" to "Acme Production". Right after that, every personal saved filter view disappeared for the two people we've checked (Jess and Hamid). The shared views are still there. We're on 4.12.3 (the latest).
Both of them use Chrome 129 on Windows 11. We haven't tried other browsers or tried renaming it back.
Jess says she had about 12 saved views that she uses every day, so this is blocking her.

Test B output

Classification: NEW

Match: none Related-but-different: OB-3310 (4.12.2)
Reasoning: No changelog entry covers personal saved filter views disappearing after a workspace rename. OB-3310 concerns saved filters not being *applied* when opening a shared dashboard link. That is a different symptom (views are present but not applied) and a different trigger (a shared link, not a rename), so it isn't a match.

Engineering bug report

Title: Saved filter views — personal views disappear for users (after workspace rename "Acme-Prod" -> "Acme Production")
Severity: S2 major — Personal saved views are lost for at least two users, and one of them (Jess, about 12 daily-use views) says she is blocked. Shared views are unaffected and the scope is not yet known to be wider.
Environment: Product 4.12.3; Chrome 129 on Windows 11 (Jess and Hamid); Enterprise plan; workspace "Acme Production" (formerly "Acme-Prod"). Ticket T-71044.
Summary: After an admin renamed the workspace, the personal saved filter views for Jess and Hamid disappeared. Shared views are still present.

Steps to reproduce:

1. In a workspace named "Acme-Prod", have users with personal saved filter views. (inferred)
2. An admin renames the workspace to "Acme Production" (done this morning, per the customer).
3. Jess and Hamid open their saved filter views. (inferred)

Expected: Personal saved views are kept after a workspace rename.
Actual: Personal saved views for both checked users are gone. Shared views remain.
Frequency / scope: Unknown if it always happens. 2 of 2 users checked are affected, and the customer has not checked anyone else.
Workaround: None known. The customer has not tried renaming back or using other browsers.
Changelog check: Searched 4.12.3 back to 4.11.2 and the open known issues for saved filters/views, workspace rename and personal-versus-shared views. Related-but-different: OB-3310 (4.12.2), as above. There are no matching open known issues.

NEEDS INFO from customer:

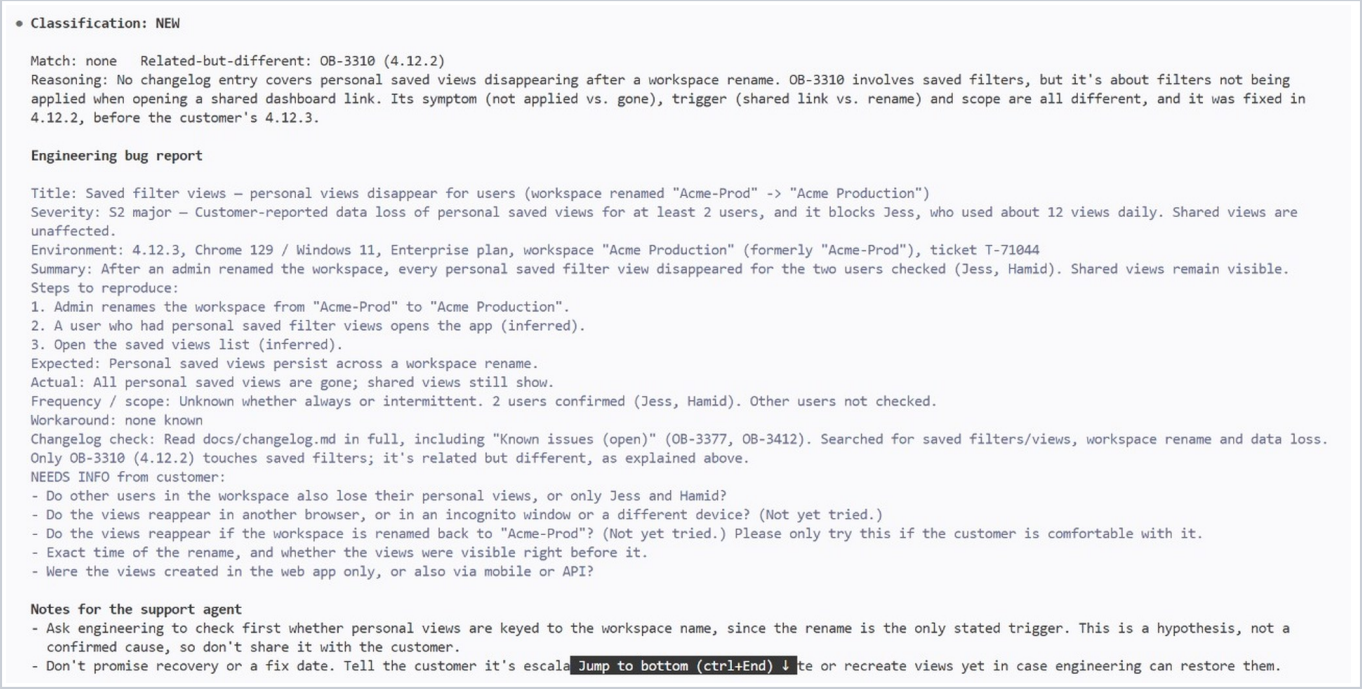
- Exact time of the rename, and whether the views vanished immediately or only after the next login or refresh.
- Are the views gone for every user in the workspace, or only for some (beyond Jess and Hamid)?
- Do the views still appear in a different browser or in the mobile app, or after signing out and back in?
- Has the workspace been renamed before, and did any other admin changes happen around the same time?
- Is there any error message or banner when opening the views list?
- Are they willing to rename the workspace back to "Acme-Prod" as a test, if engineering asks for it?

Notes for the support agent

- Don't tell the customer to rename the workspace back until engineering confirms it's safe. It could affect data, and it is unconfirmed whether it would restore the views.
- Tell Jess that engineering is investigating. Don't promise recovery or a timeline, since none is known.
- The customer only checked two users. Ask about the rest of the workspace so engineering has the real scope.

Result: NEW; OB-3310 rejected as a match; unstated repro steps marked as inferred; open questions listed under NEEDS INFO (six in the recorded run, five in the live run below).

Screenshot, live run of Test B: classification, reasoning, the engineering bug report, the questions for the customer and notes for the support agent



```
• Classification: NEW

  Match: none   Related-but-different: OB-3310 (4.12.2)
  Reasoning: No changelog entry covers personal saved views disappearing after a workspace rename. OB-3310 involves saved filters, but it's about filters not being
  applied when opening a shared dashboard link. Its symptom (not applied vs. gone), trigger (shared link vs. rename) and scope are all different, and it was fixed in
  4.12.2, before the customer's 4.12.3.

  Engineering bug report

  Title: Saved filter views — personal views disappear for users (workspace renamed "Acme-Prod" -> "Acme Production")
  Severity: S2 major — Customer-reported data loss of personal saved views for at least 2 users, and it blocks Jess, who used about 12 views daily. Shared views are
  unaffected.
  Environment: 4.12.3, Chrome 129 / Windows 11, Enterprise plan, workspace "Acme Production" (formerly "Acme-Prod"), ticket T-71044
  Summary: After an admin renamed the workspace, every personal saved filter view disappeared for the two users checked (Jess, Hamid). Shared views remain visible.
  Steps to reproduce:
  1. Admin renames the workspace from "Acme-Prod" to "Acme Production".
  2. A user who had personal saved filter views opens the app (inferred).
  3. Open the saved views list (inferred).
  Expected: Personal saved views persist across a workspace rename.
  Actual: All personal saved views are gone; shared views still show.
  Frequency / scope: Unknown whether always or intermittent. 2 users confirmed (Jess, Hamid). Other users not checked.
  Workaround: none known
  Changelog check: Read docs/changelog.md in full, including "Known issues (open)" (OB-3377, OB-3412). Searched for saved filters/views, workspace rename and data loss.
  Only OB-3310 (4.12.2) touches saved filters; it's related but different, as explained above.
  NEEDS INFO from customer:
  - Do other users in the workspace also lose their personal views, or only Jess and Hamid?
  - Do the views reappear in another browser, or in an incognito window or a different device? (Not yet tried.)
  - Do the views reappear if the workspace is renamed back to "Acme-Prod"? (Not yet tried.) Please only try this if the customer is comfortable with it.
  - Exact time of the rename, and whether the views were visible right before it.
  - Were the views created in the web app only, or also via mobile or API?

  Notes for the support agent
  - Ask engineering to check first whether personal views are keyed to the workspace name, since the rename is the only stated trigger. This is a hypothesis, not a
    confirmed cause, so don't share it with the customer.
  - Don't promise recovery or a fix date. Tell the customer it's escala Jump to bottom (ctrl+End) ↓ te or recreate views yet in case engineering can restore them.
```

Live run: Classification NEW, no match, with OB-3310 named as related but different and the reason given. Severity S2; steps the customer did not state are marked '(inferred)'; the changelog check names the known issues it read. The two notes for the support agent say not to promise recovery or a fix date. A 'Jump to bottom' tooltip from the terminal overlaps the last line; the full line is in `tests/bug-report-writer/output_B_new_bug.md`.

4.3 Agent: at-risk-briefing

Prompt (run with `--agent at-risk-briefing`): "Prepare the pre-call briefing for customer cust_1042 (Helix Dental Group). Today is 2026-10-03." The mock sources hold three customers, including a look-alike, **Helix Dental Supply (cust_1043)**, whose ticket says it wants to cancel. That record must not leak into the briefing. Planted facts for cust_1042: 40 seats, renewal 2026-11-20, weekly active users 31 to 12, three tickets on the same reports bug, an open ticket unanswered since 09-12, 'evaluating alternatives', a pending 40 to 25 seat cut and one failed payment (INV-8841).

Output (agent tools: Read, Grep, Glob only)

At-Risk Briefing — Helix Dental Group (cust_1042)

Prepared: 2026-10-03 | Renewal: 2026-11-20 (48 days) | Plan: Pro, annual, 40 seats, ACV $19,600 (+ Advanced Reporting add-on, $150/month)

Snapshot

Dental practice group and customer since 2024-11-20. Contacts: Dr. Marcus Webb (admin champion), Priya Nair (billing).

Risk signals (most severe first)

1. Customer says they are evaluating alternatives and must decide before renewal. Their 09-20 and 09-27 messages have had no reply (last support reply 09-12, so 6 days unanswered). [tickets: T-51733]
2. Scheduled reports have failed three times in three months (July, August, September). T-51733 is still open and high priority. Managers have gone back to spreadsheets. CSAT was 3, then 2. [tickets: T-50871, T-51240, T-51733]
3. Seat use fell from 77.5% (31/40) to 30% (12/40) between 08-10 and 09-28. Weekly active users are down 61%. [usage: cust_1042]
4. Reports run fell 410 to 95 (-77%). Scheduled reports delivered fell 52 to 6 (-88%). [usage: cust_1042]
5. A seat cut from 40 to 25 at renewal has been pending since 09-02. The customer asked about it again on 09-25, and the only reply says it was forwarded to billing. [billing: pending_changes; tickets: T-51902]

What's still working

- Invoices are paid. The 08-14 card decline was retried and paid on 08-19. [billing: INV-8841]
- Admin last logged in 09-30, so the champion is still engaged. [usage]

Likely churn driver (Inference)

Inference: the recurring scheduled-report failure, combined with slow support replies, has eroded trust and use. That is driving both the seat cut and the competitor evaluation.

Recommended for the call

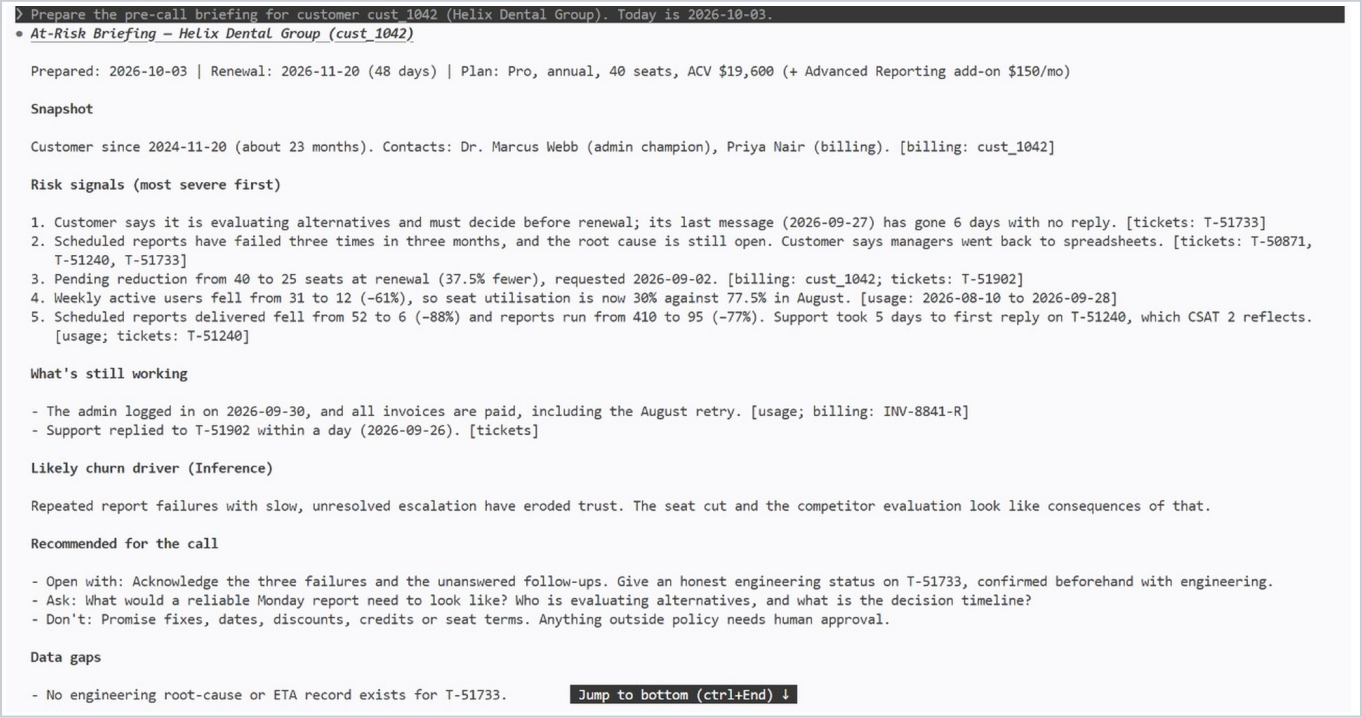
- Open with: Acknowledge the third report failure and the unanswered 09-20 and 09-27 messages. Give the current engineering status of T-51733, confirmed beforehand.
- Ask: What would a reliable reporting fix need to look like? Which teams still use Orbitly? Which alternatives, and what decision timeline?
- Don't: Promise a fix date, discounts, credits or seat terms. Don't blame the customer for the lower usage.

Data gaps

- No engineering root-cause or ETA is in the data.
- No billing reply on the seat reduction or invoice copy. Seat-cut request date differs (billing 09-02, ticket 09-25).
- Unclear whether the ACV includes the add-on. No 25-seat price is given.

Result: 375 words; renewal in 48 days; seat use 77.5% to 30%; three tickets on one bug; unanswered messages flagged; nothing from cust_1043.

Screenshot, live run: the pre-call briefing for Helix Dental Group (cust_1042), as the account manager reads it



Live run: renewal in 48 days; weekly active users 31 to 12 (seat use 77.5% to 30%); three report failures; a 40 to 25 seat reduction pending; the customer's last message unanswered for 6 days. Every fact carries a source tag, the churn driver is labelled as inference, the call guidance includes a 'Don't' line, and the one data gap (no engineering root cause or ETA for T-51733) is stated instead of guessed. Nothing from the look-alike customer appears.

4.4 Agent: ticket-quality-review

Prompt: "Run the weekly quality review on `data/closed-tickets/closed_sample_2026-09-21.md`." The file holds eight closed tickets with planted flaws: wrong refund window, wrong prices, unapproved discount, banned phrases, unanswered question, empty hand-off and two clean tickets. The answer key is `tests/ticket-quality-review/ground_truth_NOT_shown_to_agent.md`.

Input excerpt (all eight tickets are in the repo; first three shown)

Weekly sample — closed tickets (week of 2026-09-21)

Reference date for windows: closure date of each ticket.

T-60101 (closed: resolved)

Customer (Aiko, Starter monthly): Our invite emails never reach one of our contractors. Can you help?
Support (Sam): Hi Aiko, thanks for flagging this. Invite emails could land in spam for some providers; we fixed that in a recent release. Ask your contractor to check spam and mark us as safe, then update to the latest version. I'll check back with you on Thursday to confirm it arrived. Sam / Orbitly Support
Closure: Customer replied "Got it, thanks" on Thursday.

T-60102 (closed: resolved)

Customer (Rob, Pro annual, bought 45 days ago): I'd like a refund on our annual plan, we barely use it.
Support (Chris): Hi Rob, no problem at all! I've processed a full refund of $5,880 for you, it will land in 5-7 days. Chris / Orbitly Support
Closure: Refund issued by support.

T-60103 (closed: resolved)

Customer (Mei, Pro monthly): Can I get a refund for half of this month? I stopped using it on the 10th.
Support (Dev): Unfortunately that's not something we can do. As I already explained in the FAQ, partial months are never refunded and per my last email you should have cancelled earlier. Dev / Orbitly Support
Closure: Customer silent after reply, auto-closed.

Output

Weekly Ticket Quality Review

Sample: 8 closed tickets (all graded; 8 is under the 15-ticket sampling threshold) | Pass: 2 (25%) | Mean score: 5.8/10 | Auto-fails: 3

Scorecard

Ticket	A	B	C	D	E	Total	Verdict	Evidence for any score below 2
T-60101	2	2	2	2	2	10	PASS	None. Note: "we fixed that in a recent release" can't be checked against the changelog, which is outside the files I was given.
T-60102	0	0	2	1	2	5	AUTO-FAIL	A/B: the customer bought 45 days ago, but Chris wrote "I've processed a full refund of $5,880". Policy 2.1 allows only 30 days, so there is no refund. Policy 2.6 says support never quotes an amount. D: "it will land in 5-7 days" is an unapproved timeline with no owner.
T-60103	2	2	0	0	2	6	FAIL	C: "Unfortunately", "As I already explained" and "per my last email" are all banned phrases. "you should have cancelled earlier" blames the customer. D: the reply ends with no next step and offers no alternative.
T-60104	0	2	1	0	0	3	FAIL	A/E: the customer asked how to export all project data to CSV. The reply only says "You can invite your auditors as view-only guests", so the question is ignored. C: "thanks for reaching out" is filler. D: "Let me know if you need anything else!" is not a next step.
T-60105	0	0	2	1	1	4	AUTO-FAIL	A/B: "Pro is $39 per seat per month… $390 per seat per year". Policy 1 gives $49 and $490. D: "Happy to help with anything else" has no next step or owner. E: the customer asked what the annual saving is. The reply says only "cheaper" and gives no saving.
T-60106	0	2	1	0	0	3	FAIL	A/E: the customer asked "Is this a known problem?" and the reply never answers it. C: "Engineering is looking into it" puts the action on another team instead of owning it. D: it is an empty hand-off, and the closure notes "no owner, reference or follow-up".
T-60107	2	2	2	2	2	10	PASS	None. The 60-day window matches policy 2.4, no amount is quoted, and Billing is named as owner.
T-60108	0	0	2	1	2	5	AUTO-FAIL	A/B: "I can give you 30% off your renewal". Policy 5.2 says there are no other discounts and support must route requests to Sales. D: "just reply to confirm" is a clear step, but it commits to an unapproved concession.

Criterion averages

A 0.8 | B 1.3 | C 1.5 | D 0.9 | E 1.4

Recurring failure patterns (2+ tickets)

1. Promises refunds, discounts or prices that policy doesn't allow, or quotes wrong figures — T-60102, T-60105, T-60108 — "I've processed a full refund of $5,880", "Pro is $39 per seat per month", "I can give you 30% off your renewal" — Each one is an auto-fail.

T-60102 and T-60108 gave the customer real money. T-60105 quoted a price that doesn't match the policy. Policy 2.6 and 5.2 say support must not quote amounts or offer custom discounts. These cases need Billing or Sales.

2. Customer's actual question not answered — T-60104, T-60106 (also partly T-60105) — "You can invite your auditors as view-only guests" (the question was about CSV export); "Engineering is looking into it" (the question was whether it is a known problem) — The customer has to write back, and the ticket is still closed as resolved.
3. No clear next step or owner at the end of the reply — T-60103, T-60104, T-60105, T-60106 — "Let me know if you need anything else!", "Happy to help with anything else", "Engineering is looking into it" — Generic sign-offs and vague hand-offs leave the customer with nothing to act on. Nobody owns the issue afterward.

One-offs

- T-60103: Banned phrases and blaming the customer ("Unfortunately", "As I already explained… per my last email you should have cancelled earlier"). The policy answer itself was correct.
- T-60102: Gave an unapproved processing timeline ("5-7 days") on top of the wrong refund.
- T-60101: Passed. The claim that the issue was fixed in a recent release wasn't verified against the changelog.

Coaching recommendations

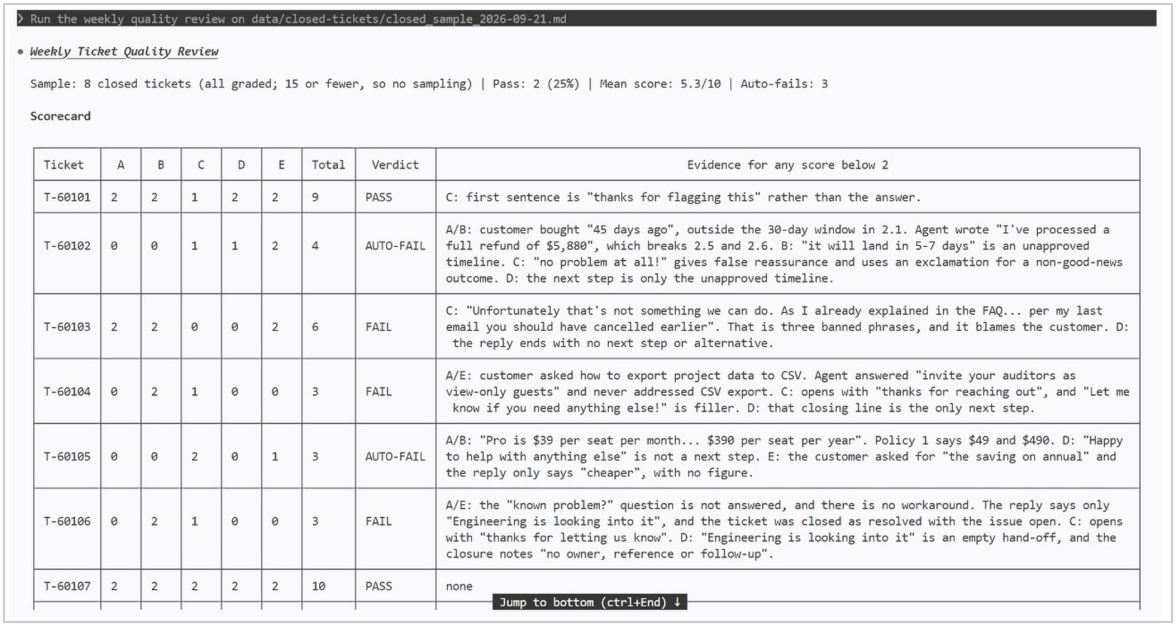
1. (Pattern 1) For any refund, price or discount reply, check the figure and the window in `docs/policy.md` first, and don't quote an amount or timeline. Send discount or price-match requests to Sales (policy 5.2). Send anything outside policy to the Billing Lead, and say plainly that it needs approval. Chris and Dev should review T-60102, T-60105 and T-60108, and Billing should look at the refund and the discount already granted.
2. (Pattern 2) Before closing, reread the customer's message and confirm each question has an answer. If the answer isn't known, say so and say who will find out. Don't close as "resolved" without an answer. Don't use "escalated to engineering" without a reference or owner.
3. (Pattern 3) End every reply with one next step and a named owner, for example "I'll update you by …" or "Billing will confirm by email". Don't use "Let me know if you need anything else".
4. (One-off T-60103) Remind Dev of the banned-phrase list and the "own it, don't blame" principle. The no-partial-refund answer was right and only the wording needs work. He could add what we can offer instead, such as cancelling before the next cycle.

Not gradable

- none

Result: eight tickets graded; 2 pass, 3 auto-fails (T-60102, T-60105, T-60108), mean 5.8/10 in the recorded run; three recurring patterns with four coaching actions. The live run below graded the same file at 5.3/10 mean with the same split (2 pass, 3 auto-fails).

Screenshot, live run: header and scorecard with evidence for every score below 2 (first screen of the report)



> Run the weekly quality review on data/closed-tickets/closed_sample_2026-09-21.md

• *Weekly Ticket Quality Review*

Sample: 8 closed tickets (all graded; 15 or fewer, so no sampling) | Pass: 2 (25%) | Mean score: 5.3/10 | Auto-fails: 3

Scorecard

Ticket	A	B	C	D	E	Total	Verdict	Evidence for any score below 2
T-60101	2	2	1	2	2	9	PASS	C: first sentence is "thanks for flagging this" rather than the answer.
T-60102	0	0	1	1	2	4	AUTO-FAIL	A/B: customer bought "45 days ago", outside the 30-day window in 2.1. Agent wrote "I've processed a full refund of $5,880", which breaks 2.5 and 2.6. B: "it will land in 5-7 days" is an unapproved timeline. C: "no problem at all!" gives false reassurance and uses an exclamation for a non-good-news outcome. D: the next step is only the unapproved timeline.
T-60103	2	2	0	0	2	6	FAIL	C: "Unfortunately that's not something we can do. As I already explained in the FAQ... per my last email you should have cancelled earlier". That is three banned phrases, and it blames the customer. D: the reply ends with no next step or alternative.
T-60104	0	2	1	0	0	3	FAIL	A/E: customer asked how to export project data to CSV. Agent answered "invite your auditors as view-only guests" and never addressed CSV export. C: opens with "thanks for reaching out", and "Let me know if you need anything else!" is filler. D: that closing line is the only next step.
T-60105	0	0	2	0	1	3	AUTO-FAIL	A/B: "Pro is $39 per seat per month... $390 per seat per year". Policy 1 says $49 and $490. D: "Happy to help with anything else" is not a next step. E: the customer asked for "the saving on annual" and the reply only says "cheaper", with no figure.
T-60106	0	2	1	0	0	3	FAIL	A/E: the "known problem?" question is not answered, and there is no workaround. The reply says only "Engineering is looking into it", and the ticket was closed as resolved with the issue open. C: opens with "thanks for letting us know". D: "Engineering is looking into it" is an empty hand-off, and the closure notes "no owner, reference or follow-up".
T-60107	2	2	2	2	2	10	PASS	none

Jump to bottom (ctrl+End) ↓

Shows the header (8 tickets, 2 pass = 25%, mean 5.3/10, 3 auto-fails) and rows T-60101 to T-60107; the last row, patterns and coaching sections continue below the visible screen (full report: `tests/ticket-quality-review/output_week_2026-09-21.md`).

5. Verification

`tests/verify_outputs.py` checks the captured outputs without any LLM: it re-runs the figure checker, recomputes the QA scorecard arithmetic, compares against the planted answer key and checks scope. Results are in `tests/verification_report.txt`.

Component	Checks passed	What is checked
reply-drafter (2 runs)	12 / 12	Figures only from the policy (script), no banned phrases, customer figures not echoed, routing to Sales, refund decision follows the live window (30 vs 45 days)
bug-report-writer (2 runs)	7 / 7	Correct class (KNOWN-FIXED vs NEW), right changelog ids, distractor not treated as a match, no write-up for a known issue, environment facts kept
at-risk-briefing agent	11 / 11	Within one page (words counted), all planted facts present, inference labelled, no data from the look-alike customer
ticket-quality-review agent	13 / 13	Row totals, averages, mean and pass rate recomputed; the three auto-fails, the two passes and the tone, accuracy and ownership flaws found; patterns grouped correctly
agent tool scope (read-only)	2 / 2	Frontmatter `tools:` contains only Read, Grep, Glob
project scope (no orchestration)	1 / 1	No orchestrator, coordinator or pipeline files exist
Total	**46 / 46**	

Defects found in testing and fixes

Finding	Fix
The first reply run could not load the skill natively because headless permission for the Skill tool was missing; the model read the file by hand. That run cost about $0.31 against about $0.12 once fixed.	Skill tool allowed in the runner; all skill tests use native skill invocation.
The figure checker flagged a draft that repeated the customer's '40%', and the skill stalled because it could not edit its draft.	Edit added to the skill's allowed tools. The guardrail worked as designed.
The first briefing ran 587 words, then 486, against a 400-word one-page limit; models count words poorly.	Word-count instruction replaced with structural limits per section; final briefing is 375 words.
Reply drafts used markdown bold and described what Sales 'will' do, which promises another team's action.	Plain-text email rule added, plus a rule that hand-offs describe the writer's own action.
The QA agent wrapped its report in a code fence and wrote 'FAIL (auto-fail)' in the verdict cell.	Output spec now forbids the fence and fixes the verdict wording.
Two verifier patterns rejected valid output: a 'full cash refund' wording and a verdict format.	Verifier patterns corrected; outputs unchanged.

6. AI economics: what each run cost

Cost per run is input tokens plus output tokens, with retries counted. Figures are the list-price cost reported by the CLI for one run of each final version on claude-sonnet-5-5, covering the whole agentic loop (file reads and tool turns), not only the final text. Metadata is saved in `tests/*/run_*.json`.

Component	Test	Cost per run (USD)	Turns	Input-side tokens	Output tokens
at-risk-briefing	cust_1042	0.056	5	26,543	1,882
bug-report-writer	A_known_fixed	0.087	4	121,654	1,063
bug-report-writer	B_new_bug	0.091	4	121,937	1,394
reply-drafter	1_policy_v2026.10	0.119	7	166,222	2,493
reply-drafter	2_policy_changed_to_45_days	0.129	9	210,162	2,509
ticket-quality-review	week_2026-09-21	0.082	5	18,813	4,315

Unit cost. A drafted reply costs about $0.12, a briefing about $0.06 and an eight-ticket QA batch about $0.08. Scaled linearly that is roughly $124 per 1,000 drafted replies. Regenerations multiply it, so budget one, two and three generations per ticket as the conservative, realistic and aggressive cases. Infrastructure, staff and ticketing-tool costs are out of scope.

Pick: stay on a Sonnet-class model for all four components until an A/B test shows a Haiku-class model holds the regeneration rate. A cheaper model lowers cost per call but can raise total cost if weaker first drafts force more regenerations. The test measures regeneration rate, cost per accepted reply and agent satisfaction, and has not been run. The two agents are the next candidates because they read the most files.

Where tokens were saved. Skills and agents read shared documents instead of embedding them, and the deterministic checks (figure checker, verifier arithmetic) cost no tokens. Capability 1 (Neither) costs no model tokens at all, which is itself the economic decision: a deterministic rule is cheaper, faster and safer than an LLM for keyword escalation.

Business case. The costs sit on the support side, so the case is internal: AI cost per ticket against agent minutes saved and refund or pricing errors avoided.

7. Limits and next steps

- **Mock data.** The three agent sources are files (`data/billing`, `data/usage`, `data/tickets`). In production the same agents would use read-only integrations for billing, product analytics and the ticketing tool, and the `tools:` list would name those specific tools instead of file readers. Scope and instructions would not change.
- **Out of scope by design.** No orchestration, coordinator, end-to-end wiring or pipeline. Capability 1 (Neither) is a documented design decision, not code.
- **Non-determinism.** LLM outputs vary from run to run, so verification is deterministic and re-runnable (`python3 tests/verify_outputs.py`) and the runner can regenerate every output (`bash tests/run_tests.sh`).
- **Fictional policy and rubric.** Both were written to create realistic traps: refund windows, price points, a look-alike customer and a similar-sounding changelog entry.
- **Enforcement.** `check_figures.py` runs because the skill's instructions say to run it, so it is a step Claude follows. Anthropic's documentation says hooks always fire on their event while instructions can be interpreted. **Pick: wire the figure check as a hook before any production use**, so a draft cannot go out without it.

8. Submission contents and sources

Deliverable	Where
PDF: mapping, justifications, standalone test evidence	This file
Skills and agents built	`.claude/skills/reply-drafter`, `.claude/skills/bug-report-writer`, `.claude/agents/at-risk-briefing.md`, `.claude/agents/ticket-quality-review.md`
Test inputs and outputs	`tests/` (inputs, `output_*.md`, `run_*.json`, `draft_*.md`), `data/`, `docs/`
Screenshots of live runs	`screenshots/` (one per component, section 4)
Reproduce and verify	`bash tests/run_tests.sh`, `python3 tests/verify_outputs.py`

Sources

- Extend Claude Code (Anthropic): skills vs subagents vs CLAUDE.md vs hooks. Subagents run in isolated context and return summaries; CLAUDE.md under 200 lines; instructions are requests, hooks are enforcement.
- Skill authoring best practices (Anthropic). Pre-made scripts for deterministic operations; match specificity to task fragility.